

Module-5

Introduction to Groups Theory

Groups:-

Let G be a non-empty set and $*$ be a binary operation on G . Then G is called a group. Then G is called a group under the operation $*$ if the following conditions are satisfied:

- 1) $a * b \in G \quad \forall a, b \in G$. [Closure]
- 2) $(a * b) * c = a * (b * c) \quad \forall a, b, c \in G$ (Associative).
- 3) $\exists e \in G$ such that $a * e = e * a = a, \forall a \in G$. [e is called identity element under $*$].
- 4) For every $a \in G$, there is an element $a' \in G$ such that $a * a' = a' * a = e$. (Here a' is called an inverse of a under $*$ and is usually denoted by a^{-1})

A group G under $*$ will be denoted by $(G, *)$.

$\Rightarrow$ A group G is said to be abelian if it satisfies commutative law. i.e. $a * b = b * a \quad \forall a, b \in G$.

* A group $(G, *)$ is said to be a finite group of order n if G is a finite set with $|G| = n$.

$O(G) = n$ (order of G is n .)

* A group $(G, *)$ where G is not a finite set, is called an infinite group.

Eg: $(\mathbb{Z}, +)$ is ^{abelian} group.

Sol. (*) WKT. the set $\mathbb{Z}$ of all integers is closed under addition.

$$\text{i.e. } \forall a, b \in \mathbb{Z}, \quad a+b \in \mathbb{Z}.$$

(*) WKT the set $\mathbb{Z}$ of all integers is associative under addition

$$\text{i.e. } \forall a, b, c \in \mathbb{Z}, \quad a+(b+c) = (a+b)+c.$$

$$(*) \text{ WKT } a+0 = 0+a = a \quad \forall a \in \mathbb{Z}.$$

$\therefore 0$ is an identity element in $\mathbb{Z}$.

(*) $\forall a \in \mathbb{Z}, \exists -a \in \mathbb{Z}$, such that

$$a+(-a) = (-a)+a = 0.$$

$$\text{i.e. } a^{-1} = -a \text{ and } (-a)^{-1} = a.$$

$\therefore$ Inverse exists $\forall a \in \mathbb{Z}$.

$\therefore (\mathbb{Z}, +)$ is a group.

$$* \quad \forall a, b \in \mathbb{Z}, \quad a+b = b+a.$$

$\therefore (\mathbb{Z}, +)$ is an abelian group.

Eg: $(\mathbb{Z}, -)$ is not a group,

(Associative law not satisfied).

Problems;.

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1. ~~Let~~ If G be a set of all non-zero real numbers and let $a * b = \frac{ab}{2}$ then show that $(G, *)$ is an abelian group.

Sol. Given $a * b = \frac{ab}{2}$, $a, b \in R_0 = G$.

$$(*) \forall a, b \in G, \text{ then } a * b = \frac{ab}{2} \in G$$

$\therefore$ Closure law is satisfied.

$$(*) \forall a, b, c \in G,$$

$$(a * b) * c = \left(\frac{ab}{2} \right) * c = \frac{abc}{4}.$$

$$\text{and } a * (b * c) = a * \left(\frac{bc}{2} \right) = \frac{abc}{4}.$$

$$\therefore (a * b) * c = a * (b * c).$$

$\therefore$ Associative law is satisfied,

$$(*) \forall a \in G, a * e = e * a = a.$$

$$\frac{ae}{2} = \frac{ea}{2} = a$$

$$\frac{ae}{2} = a \Rightarrow \boxed{e=2}, e \in G.$$

$\therefore 2$ is the identity element in G

(*) Inverse

for any $a \in G$,

$$a * a^{-1} = e$$

$$\frac{aa^{-1}}{2} = 2$$

$$aa^{-1} = 4.$$

$$a^{-1} = \frac{4}{a} \in G.$$

$\therefore$ Every a in G has $a^{-1} = \frac{4}{a}$ as the inverse in G under $*$.

(*) Commutative law

$\forall a, b \in G$,

$$a * b = \frac{ab}{2} = \frac{ba}{2} = b * a$$

$\therefore$ Commutative law is satisfied.

$\therefore (G, *)$ is Abelian.

$$a \oplus b = \frac{3ab}{2}$$

closure law

$$\forall a, b \in G, a \oplus b = \frac{3ab}{2} \in G$$

$\therefore (G, \oplus)$ is

Associative law

$$\forall a, b, c \in G$$

$$a \oplus (b \oplus c) = (a \oplus b) \oplus c$$

$$a \oplus \left[\frac{3bc}{2} \right] = \frac{3ab \cdot c}{2}$$

$$\frac{3 \cdot a \cdot \frac{3bc}{2}}{2}$$

$$\frac{3 \cdot \frac{3ab}{2} \cdot c}{2}$$

$$\frac{9abc}{4}$$

$$\frac{9abc}{4}$$

Commutative

(2) For the set of non-zero real numbers $(G, \oplus)$ is Abelian (5)
 s.t. $a \oplus b = \frac{3ab}{2}$ is abelian
 if $a \oplus b = \frac{3ab}{2}$
 and hence $b \oplus a = \frac{3ba}{2}$

Identity $2^1, 4^1$

$$\forall a \in G \exists e \in G$$

$$\Rightarrow a \oplus e = e \oplus a = a$$

$$a \oplus e = a$$

$$\frac{3ae}{2} = a$$

$$e = \frac{2}{3}$$

is the identity element

Inverse law

$$\forall a \in G \exists a^{-1} \in G$$

$$a \oplus a^{-1} = a^{-1} \oplus a = e$$

$$\Rightarrow a \oplus a^{-1} = e$$

$$\frac{3aa^{-1}}{2} = \frac{2}{3}$$

$$aa^{-1} = \frac{4}{9}$$

$$a^{-1} = \frac{4}{9a}$$

$$\frac{4}{9a} \cdot a = \frac{4}{9}, \quad 3^{-1} = \frac{4}{27}$$

$1, \omega, \omega^2$
closure law

Table

$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

$$A = \{1, \omega, \omega^2\}$$

$$\omega^3 = 1$$

$$\omega^4$$

3) If $\circ$ is an operation on $\mathbb{Z}$ defined by $x \circ y = x + y + 1$, prove that $(\mathbb{Z}, \circ)$ is an abelian group.

← previous

(7)

2) Show that $(A, \circ)$ is an abelian group where

$$A = \{a \in \mathbb{Q} \mid a \neq -1\} \text{ and for any } a, b \in A,$$

$$a \circ b = a + b + ab.$$

Sol

$$A = \{a \in \mathbb{Q} \mid a \neq -1\}$$

$$a \circ b = a + b + ab.$$

(*) when $a \neq -1, b \neq -1,$

$$a + b + ab \neq -1.$$

$\therefore \circ$ is a binary operation on A

$\therefore$ closure law is satisfied.

Closure

$$\forall a, b \in A$$

$$a \circ b = a + b + ab \in A$$

(*) For any $a, b, c \in A$, we have

$$a \circ (b \circ c) = a \circ (b + c + bc)$$

$$= a + (b + c + bc) + a(b + c + bc)$$

$$a \circ (b \circ c) = a + b + c + bc + ab + ac + abc.$$

$$\text{and } (a \circ b) \circ c = (a + b + ab) \circ c$$

$$= (a + b + ab) + c + (a + b + ab)c.$$

$$(a \circ b) \circ c = a + b + c + ab + ac + bc + abc.$$

$$\therefore a \circ (b \circ c) = (a \circ b) \circ c.$$

Associative law is satisfied.

(*) For any $a \in A$, we have

$$a \circ 0 = a + 0 + a \cdot 0 = a$$

$$0 \circ a = 0 + a + 0 \cdot a = a$$

$\therefore 0$ is the identity element in A

(*) For any $a \in A$,

$$a \circ a^{-1} = 0.$$

$$a + a^{-1} + a a^{-1} = 0.$$

$$a + a^{-1}(1+a) = 0.$$

$$a^{-1} = \frac{-a}{1+a} \in A.$$

$\therefore$ Inverse exists for every $a \in A$.

$\therefore (A, \circ)$ is a group.

(*) $\forall a, b \in A$,

$$\begin{aligned} a \circ b &= a + b + ab \\ &= b + a + ba \\ &= b \circ a. \end{aligned}$$

$\therefore$ Commutative law is satisfied.

$\therefore (A, \circ)$ is an Abelian group.

Define group.

37. Show that fourth roots of unity is an abelian group.

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Sol. The fourth roots of unity are

$$W_4 = \{1, -1, i, -i\}.$$

Let us consider the operation table for mutual multiplication on W_4 :

x	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

* All the elements of the table belong to W_4 .
 $\therefore$ Closure law is satisfied.

* Since multiplication is associative in complex numbers, it is associative in W_4 .

* The row headed by 1 is similar to the top most row. $\therefore$ 1 is the identity element under $\times$.

* Every element has its inverse.

$$1^{-1} = 1, \quad (-1)^{-1} = -1, \quad i^{-1} = -i, \quad (-i)^{-1} = i$$

* Multiplication is commutative.

$\therefore$ W_4 is an abelian group with order 4.

$$(x^4 - 1 = 0$$

$$((x^2)^2 - 1) = 0.$$

$$(x^2 + 1)(x^2 - 1) = 0$$

$$x^2 = -1, \quad x^2 = 1$$

$$x = \pm i, \quad x = \pm 1.$$

4) Define Abelian group. Show that cube roots of unity is an Abelian group. (10)

$$G = \{1, \omega, \omega^2\}$$

$$\omega^3 = 1$$

$$\omega^3 = 1$$

$$1, \omega, \omega^2$$

Theorem:

In a group, there exists only one identity element. (11)

(or) [Identity element in a group G is unique].

Proof:

To prove: In a group, there exists only one identity element.

Let us consider that e_1 and e_2 are identity elements in a group G .

Since e_1 is an identity element in G ,
we have $ae_1 = e_1a = a$. $\forall a \in G$.

This is true for $a = e_2$, since $e_2 \in G$.

$$\Rightarrow e_2e_1 = e_1e_2 = e_2.$$

Similarly because e_2 is an identity element in G ,

$$\text{we get } ae_2 = e_2a = a.$$

$$\text{Thus } e_1 = e_1e_2 = e_2e_1 = e_2.$$

This shows that e_1 and e_2 are same.

$\therefore G$ has only one identity element.

Theorem:

In a group G , every element has only one inverse.

(or) [Inverse of each element in a group is unique].

Let e be the identity element in G .

Suppose a' and a'' are inverses of an element $a \in G$.

$$\text{Then } a' = a'e.$$

$$= a'(aa'').$$

$$= (a'a)a''$$

$$a' = ea''$$

$$[a = a * e, \text{ for } x \in G]$$

$$[a'' \text{ is inverse of } a].$$

$$a' = a''.$$

(12)

Thus, a' and a'' cannot be different.

Hence the proof.

Note: * $e^{-1} = e$

$$* (a^{-1})^{-1} = a$$

$$* (ab)^{-1} = b^{-1}a^{-1}.$$

$$* xa = xb \Rightarrow a = b.$$

$$* ax = bx \Rightarrow a = b.$$

The Klein 4-group.

(13)

* Define Klein 4-group. And if $A = \{e, a, b, c\}$ then show that this is a Klein-4 group.

Ans Consider a set $A = \{e, a, b, c\}$. On this set, suppose we define a binary operation described by

$$\begin{aligned} a \cdot e &= e \cdot a = a & \text{and} & & a \cdot a &= e & \text{and} & & a \cdot b &= b \cdot a = c \\ b \cdot e &= e \cdot b = b & & & b \cdot b &= e & & & b \cdot c &= c \cdot b = a \\ c \cdot e &= e \cdot c = c & & & c \cdot c &= e & & & a \cdot c &= c \cdot a = b. \end{aligned}$$

$$e \cdot e = e \cdot e = e$$

If the elements satisfy the above condition, then it is Klein-4 group.

Multiplication table for V_4

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

* This is closed under binary operation.

* From the defn, we can also see that associative law is satisfied.

$$\left. \begin{aligned} (a \cdot b) \cdot c &= c \cdot c = e \\ a \cdot (b \cdot c) &= a \cdot a = e \end{aligned} \right\} \text{Associative law}$$

* We can also observe from the table that e is the identity element.

* We observe that $a \cdot a = e \Rightarrow a^{-1} = a$
 $b \cdot b = e \Rightarrow b^{-1} = b$
 $c \cdot c = e \Rightarrow c^{-1} = c$

Hence, A is a Klein-4 group.

* Inverse exist for each element.
 * Commutative law is also satisfied.

* Define Klein 4 group. Verify $A = \{1, 3, 5, 7\}$ is a Klein 4 group.

Sol Defn. as in prev. page.

$$A = \{1, 3, 5, 7\}$$

From defn, The Cayley's table becomes,

.	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

$$\left. \begin{array}{l} 1 \cdot 1 = 1, 1 \cdot 1 = 1 \\ 1 \cdot 3 = 3, 1 \cdot 3 = 3 \\ 5 \cdot 1 = 1, 5 \cdot 5 = 5 \\ 7 \cdot 1 = 1, 7 \cdot 7 = 1 \end{array} \right| \begin{array}{l} 3 \cdot 3 = 1 \\ 5 \cdot 5 = 1 \\ 7 \cdot 7 = 1 \end{array}$$

$$\begin{array}{l} \cancel{1 \cdot 3 = 3, 1 \cdot 3 = 3} \\ 3 \cdot 5 = 5, 3 \cdot 5 = 5 \\ 3 \cdot 7 = 7, 3 \cdot 7 = 7 \\ 5 \cdot 7 = 7, 5 \cdot 7 = 7 \end{array}$$

Closure law:-

We see that all the elements of the table belong to set A.

Associative law:-

We see that $\forall a, b, c \in A$,

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

$\therefore$ Associative law is satisfied.

Identity law:

1 is the identity element.

Inverse law.

$$1^{-1} = 1, 3^{-1} = 3, 5^{-1} = 5, 7^{-1} = 7$$

Inverse exists $\forall a \in A$.

Commutative law:

We see that $\forall a, b \in A$

$$a \cdot b = b \cdot a$$

$\therefore A = \{1, 3, 5, 7\}$ is a Klein 4 group.

Subgroups [SUBGROUPS]

(15)

A nonempty subset H of a group G is called a subgroup of G whenever H itself is a group under the binary operation in G .

Under the usual addition, the set of all even integers is a subgroup of the group of all integers.

Theorem 1

H is a subgroup of G if and only if, for all $a, b \in H$, we have $ab \in H$ and $a^{-1} \in H$.

Proof:

we need to proof 2 parts

First let H is a subgroup of G .

Then H is a group on its own.

$\therefore \forall a, b \in H$, $ab \in H$ and $a^{-1} \in H$ hold

Conversely,

Suppose $\forall a, b \in H$, the conditions $ab \in H$ and $a^{-1} \in H$ hold.

Then for any $a, b, c \in H$.

$\therefore H \subseteq G$, $a, b, c \in G$.

$\therefore a(bc) = (ab)c$.

Associative law holds in H .

For any $a \in H$, $a^{-1} \in H$ by given condition and

$\therefore aa^{-1} \in H$ by given condition. [since $ab \in H$

$\therefore e \in H$.

H is a group.

where $b = a^{-1}$

Since $H \subseteq G$, H is a subgroup of G .

Hence the proof.

Theorem 2:

H is a subgroup of G if and only if, for all $a, b \in H$, we have $ab^{-1} \in H$.

Proof:

Firstly, let H is a subgroup of G .

Then H is a group by itself.

For any $a, b \in H$, then $b^{-1} \in H$.

so $ab^{-1} \in H$.

[Closure property $ab^{-1} \in H$
 $a \cdot b^{-1} \in H$
 $a \cdot b^{-1} \in H$]

Conversely,

suppose for all $a, b \in H$, the condition $ab^{-1} \in H$ holds.

$\Rightarrow aa^{-1} \in H$ ($b=a$) $\leftarrow a, a^{-1} \in H$ $ab^{-1} \in H, a, b \in H$
 $aa^{-1} \in H$
 $e \in H$ Identity

$\Rightarrow e \in H$.

This condition also yields $ea^{-1} \in H$ when $a \in H$.

Thus, $a^{-1} \in H$ for every $a \in H$.

Since $b \in H$, we have $b^{-1} \in H$.

$\therefore a(b^{-1})^{-1} \in H$ for all $a, b \in H$.

$ab \in H$.

H is closed under the binary operation of G .

Since associative law holds for all elements of G , this law holds for all elements of $H \subseteq G$.

$\therefore H$ is a subgroup of G .

Hence the proof.

Note:

For any group G , $e \in G$. Hence $\{e\} \subseteq G$ and $G \subseteq G$.
 $\therefore \{e\}$ and G of a group G are called trivial subgroups. Other subgroups are called proper subgroups.

14 Prove that the intersection of two subgroups of a group is a subgroup of the group. Is the union of two subgroups of a group a subgroup of the group? (17)

Sol Let G be a group.

Let H and K be subgroups of G .

Then for any $a, b \in H \cap K$.

$$\Rightarrow a, b \in H \text{ and } a, b \in K.$$

$$\Rightarrow ab^{-1} \in H \text{ and } ab^{-1} \in K \quad | \because H \text{ and } K \text{ are subgroups.}$$

$$\Rightarrow ab^{-1} \in H \cap K.$$

$\Rightarrow H \cap K$ is a subgroup of G . (Theorem 2)

$\therefore$ Intersection of two subgroups of a group G is a subgroup of G .

Union of two subgroups of a group need not be a subgroup.

Eg! WKT $(\mathbb{Z}, +)$ is a group.

$(2\mathbb{Z}, +)$ and $(3\mathbb{Z}, +)$ are subgroups of $(\mathbb{Z}, +)$.

$$\text{But } (2\mathbb{Z} \cup 3\mathbb{Z}) = \{ \dots, -3, -2, 0, 2, 3, 4, 6, 8, 9, \dots \}$$

Consider $2, 3 \in 2\mathbb{Z} \cup 3\mathbb{Z}$

$$\text{But } 2+3=5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}.$$

Closure law is not satisfied.

$\therefore$ Union of two subgroups need not be a subgroup.

Cyclic Group

A group G is said to be cyclic if there exists an element $g \in G$ such that every element 'a' of G is called an "integral power" of g ; i.e. 'a' is of the form g^n for some integer n . Then the element g is called a generator of the group.

If g is a generator of a cyclic group G we say that g generates G or that G is generated by g .

A cyclic group G generated by g is denoted by $\langle g \rangle$.

Ex:- Consider fourth roots of unity

$$W_4 = \{1, -1, i, -i\}$$

Here the element 'i' can generate the whole set of W_4 .

$$\text{i.e. } (i)^0 = 1, (i)^2 = -1, i^3 = -i, i^4 = 1$$

$$\therefore (W_4, \cdot) \cong \langle i \rangle$$

[Here 'i' can also be the generator]

Note:- If g is a generator of a cyclic group G , then g^{-1} is also a generator of G

Define Cyclic group and show that $(G, *)$ whose multiplication table is as given below is Cyclic

*	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	c	d	e	f	a
c	c	d	e	f	a	b
d	d	e	f	a	b	c
e	e	f	a	b	c	d
f	f	a	b	c	d	e

Here $G = \{a, b, c, d, e, f\}$

From the above table the row headed by 'a' is same as the topmost row
 $\therefore$ 'a' is the identity element.

We note that $b^2 = b * b = c$

$$c^2 = c * c = e$$

$$c^3 = c^2 * c = e * c = c$$

$$c^4 = c^3 * c = c * c = e$$

$$e^1 = e * e = e$$

$$b^3 = b^2 * b = c * b = d$$

$$b^4 = b^3 * b = d * b = e$$

$$b^5 = b^4 * b = e * b = b$$

$$b^6 = b^5 * b = b * b = c$$

We note that the whole of the group G can be generated by integral powers of b .

$$\therefore (G, *) = \langle b \rangle$$

Coset Decomposition of a group (21)

Let $(G, *)$ be a group and $(H, +)$ be a subgroup of G . For any $a \in G$, let

$$a * H = \{a * h \mid h \in H\}$$

$$H * a = \{h * a \mid h \in H\}$$

Then $a * H$ is called the left coset of H wrt a in G and $H * a$ is called the right coset of H wrt a in G .

Note:- If G is abelian, then every left coset of H is also a right coset.

2) Any two left(right) cosets of a subgroup H of a group G are either disjoint or identical

3) $|H| = |aH| = |Ha|$

Eg:- Let $(G, \cdot)$ be a group where $G = \{1, -1, i, -i\}$ and $H = \{1, -1\}$

Then $a \cdot H =$ left coset $H \cdot a =$ right coset

$1 \cdot H = \{1, -1\}$ - (cosets will be

$$H \cdot 1 = \{1, -1\}$$

$-1 \cdot H = \{-1, 1\}$ for each

$$H \cdot (-1) = \{-1, 1\}$$

$i \cdot H = \{i, -i\}$ element

$$H \cdot i = \{i, -i\}$$

$-i \cdot H = \{-i, i\}$

$$H \cdot (-i) = \{-i, i\}$$

Prove that the only left coset of a subgroup H of a group G which is also a subgroup of G is H itself.

Proof: Consider a left coset aH of H and suppose that aH is a subgroup of G . Then $e \in aH$. But $e \in H = eH$. Therefore, aH and H are not mutually disjoint. Hence $aH = H$. Thus, if aH is a subgroup of G , then aH must be identical with H . This proves the required result.

Prove that only right coset of a subgroup H of a group G which is also a subgroup of G is H itself.

Proof:

Lagrange's Theorem

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If G is a finite group and H is a subgroup of G , then the order of H divides the order of G .

Proof:

G is a finite group and H is a subgroup of G ,
 $\Rightarrow H$ is also a finite.

$\therefore$ The number of cosets of H in G is finite.

Let $Ha_1, Ha_2, \dots, Ha_r$ be the distinct right cosets of H in G .

Then $G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_r$.

$$O(G) = O(Ha_1) + O(Ha_2) + \dots + O(Ha_r)$$

But wkt $O(Ha_1) = O(Ha_2) = \dots = O(Ha_r) = O(H)$.

$$\therefore O(G) = O(H) + O(H) + \dots + O(H)$$

$\underbrace{\hspace{10em}}_{r \text{ terms}}$

$$O(G) = r O(H)$$

$\therefore O(H)$ divides $O(G)$.

Hence the proof.

1) Let G be a group with subgroup H and K .

If $|G| = 660$ and $|K| = 66$ and $K \subset H \subset G$ and find the possible value for $|H|$.

Sol By Lagrange's theorem,

$$o(H) \text{ divides } o(G).$$

$$|G| = 660, \quad |K| = 66.$$

$\therefore$ By Lagrange's theorem,

Since $K \subset H \subset G$, $|H|$ must divide $|G|$

and $|K|$ must divide $|H|$.

$$|G| = r_1 |H|$$

$$|H| = r_2 |K|$$

$$\therefore 660 = r_1 |H| \quad \text{and} \quad |H| = 66 r_2.$$

$$\therefore 660 = 66 r_1 r_2$$

$$r_1 r_2 = 10.$$

$$\therefore r_1 = 2 \text{ and } r_2 = 5 \quad (\text{or})$$

$$r_1 = 5 \text{ and } r_2 = 2.$$

$$\therefore |H| = 66 \times 5 = 330$$

$$(\text{or}) |H| = 66 \times 2 = 132$$